

Title

Preoperative intranasal dexmedetomidine reduces postoperative delirium and improves recovery in elderly patients undergoing orthopedic surgery

Abstract

Background: Elderly patients undergoing major orthopedic procedures face elevated risk for postoperative delirium with significant morbidity implications. **Objective:** To determine whether preoperative intranasal dexmedetomidine reduces delirium incidence in patients aged 60+ undergoing hip or knee arthroplasty. **Methods:** A total of 232 patients were randomly assigned to receive either dexmedetomidine (100 µg) or placebo intranasally approximately 18 hours before surgery. Delirium assessment employed validated screening instruments administered at 24, 48, and 72 hours post-op. Secondary measures included sleep quality, anxiety, pain scores, and ambulation timing. **Results:** Delirium occurred in 11.2% of the dexmedetomidine group versus 23.3% in the placebo group (relative risk 0.48; 95% CI 0.27-0.86; $P = 0.019$). Sleep quality scores improved significantly with pretreatment (mean difference 2.8 points, $P = 0.004$). Preoperative and postoperative anxiety ratings were lower in the intervention arm ($P = 0.008$). Hemodynamic stability and oxygenation remained stable in both groups. **Conclusion:** Preoperative intranasal dexmedetomidine is effective and safe for delirium reduction in elderly joint surgery patients, possibly through enhancement of preoperative sleep quality.